

Natalie Queally

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Education

Ph.D., Forestry University of Wisconsin, Madison Dissertation: “Investigating functional trait compositions in response to a changing world: Imaging spectroscopy applications” Advised by: Dr. Philip Townsend	August 2024
B.A., Geography and Environmental Studies Minors: Conservation Biology, GIS&T University of California, Los Angeles	2016

Research Appointments

NASA Postdoctoral Fellow, Jet Propulsion Laboratory	Sep 2026 – Present
Postdoctoral Scholar, Department of Ecology & Evolutionary Biology, University of California, Los Angeles	Sep 2024 – Aug 2026

Publications

Braghiere, R. K., Wang, Y., Liu, K., Chadwick, K. D., Brodrick, P. G., Shiklomanov, A. N., Schneider, F.D., Ferraz, A., Zheng, T., **Queally, N.**, Townsend, P.A., Schimel, D., & Frankenberg, C. Shifting from plant functional types to traits: Enhancing biodiversity representation in Earth system models. In press: *Ecosphere*.

Saide, P.E., Wu, Y., Arnold, M., Thapa, L.H., Soja, A., Gargulinski, E., Li, F., Wiedinmyer, C., Emmons, L.K., Tang, W., Westerling, A.L., Xu, Q., Ordway, E., **Queally, N.**, & Kueppers, L. (2026). Assessing consistency in fuel consumed between activity-based wildfire emission estimates. *Geophysical Research Letters*, 53(8), e2025GL120030

Gilbert, N., **Queally, N.**, Cooper, J.C., Eyster, H.N., & Williams, P.J. (2026). Heterogeneity, productivity, and migration drive continental evenness patterns of bird assemblages. *Global Ecology and Biogeography*, 35(2), e70205.

Queally, N., Zheng, T., Ye, Z., Kovach, K. R., Pavlick, R., Shafron, E., Schneider, F.D., & Townsend, P. A. (2025). Functional traits from imaging spectroscopy inform patterns of forest mortality during Sierra Nevada drought. *Global Change Biology*, 31(5), e70246.

Chadwick, K. D., Davis, F.W., Miner, K.R., Pavlick, R., Reynolds, M., Townsend, P. A., ..., **Queally, N.**, ..., & Schimel, D. (2025). Unlocking ecological insights from sub-seasonal visible-to-shortwave infrared imaging spectroscopy: The SHIFT campaign. *Ecosphere*, 16(3), e70194.

Greenberg, E., Thompson, D. R., Jensen, D., Townsend, P. A., **Queally, N.**, Chlus, A., Fichot, C.G., Harringmeyer, J.P., & Simard, M. (2022). An improved scheme for correcting remote spectral surface reflectance simultaneously for terrestrial BRDF and water-surface sunglint in coastal environments. *Journal of Geophysical Research: Biogeosciences*, 127(3), e2021JG006712.

Queally, N., Ye, Z., Zheng, T., Chlus, A., Schneider, F., Pavlick, R. P., & Townsend, P. A. (2022). FlexBRDF: A flexible BRDF correction for grouped processing of airborne imaging spectroscopy flightlines. *Journal of Geophysical Research: Biogeosciences*, 127(1), e2021JG006622.

In review

Berman, L., Schneider, F.D., Pavlick, R., Peery, M.Z., Wood, C.M., Zheng, T., Shafron, E., Ye, Z., **Queally, N.**, Dean, M., Winiarski, J.M., Kramer, A., & Townsend, P.A. The predictive power of new remote sensing data for biological distributions. In review: *Ecosphere*.

Shafron, E., Zheng, T., Ye, Z., Schneider, F.D., **Queally, N.**, Kovach, K.R., Pavlick, R., Townsend, P.A. The AVIRIS-Classic Western Diversity Time Series: Corrected imaging spectroscopy surface reflectance and plant trait maps. In review: *Scientific Data*.

Ongjoco, A.J., Anderegg, L.L., Vinod, N., **Queally, N.**, Sack, L., Chadwick, K.D., Brodrick, P.G., & Ordway, E.M. Seasonal changes in canopy water content provide early drought stress warning in oaks. In review: *Journal of Geophysical Research: Biogeosciences*.

Berman, L., Schneider, F.D., Pavlick, R., Peery, M.Z., Wood, C.M., Zheng, T., Shafron, E., Ye, Z., **Queally, N.**, Dean, M., Tagliabue, G., Winiarski, J.M., Kramer, A., & Townsend, P.A. Remote sensing and bioacoustics reveal avian niche partitioning and habitat filtering. In review: *Oikos*.

In preparation

Queally, N., Johny, M., Zheng, T., Chadwick, K.D., & Townsend, P.A. High frequency imaging spectroscopy data show phenological changes in trait expression in grazed landscapes. In preparation: *Ecological Applications*.

Data Publications

Zheng, T., **Queally, N.**, Ade, C., Brodrick, P. G., Chadwick, K. D., & Townsend, P. A. (2025). *SHIFT: AVIRIS-NG Derived Plant Functional Trait Mosaics*. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAAC/2453>

Zheng, T., Shafron, E., Ye, Z., Schneider, F. D., **Queally, N.**, Kovach, K. R., Pavlick, R., & Townsend, P. A. (2025). *WDTS: AVIRIS-Classic Derived Plant Trait Mosaics, 2013-2018*. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAAC/2403>

Queally, N., Davis, F.W., Chadwick, K.D., Ade, C., Anderegg, L., Angel, Y., ... & Schimel, D.S. (2024). *SHIFT: Vegetation Plot Characterization, Santa Barbara County, CA 2022*. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAAC/2295>

Queally, N., Davis, F.W., Chadwick, K.D., Ade, C., Anderegg, L., Angel, Y., ... & Schimel, D.S. (2024). *SHIFT: Vegetation Plot Photos, Santa Barbara County, CA 2022*. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAAAC/2334>

Fellowships and Awards

NASA Postdoctoral Program Fellowship (NPP) (\$185,000)	2026 – 2028
Future Investigators in NASA Earth and Space Science and Technology (FINESST) (\$100,000)	2022 – 2024
Student Research Grants Competition (\$3000 combined)	2022, 2023
Joon Lee Award (for outstanding paper in F&WE Department) (\$1000)	2022
National Science Foundation Graduate Research Fellowship Program (GRFP) (\$138,000)	2018 – 2021

Other Research Experience

Research Scientist, NASA Jet Propulsion Laboratory	Jan 2018 – Aug 2018
NASA DEVELOP Intern, NASA Jet Propulsion Laboratory	Sep 2016 – Nov 2017

Teaching

Co-Instructor, Forest Ecology UW-Madison	Fall 2022
Instructor, Forest Ecology Lab UW-Madison	Fall 2022
Guest Lecturer, Environmental Sensing Technologies UW-Madison	Spring 2020

Trainings

Queally, N. Spatially correlated functional ANOVA (scFANOVA) for comparing groups of time series. (2024). *NSF Biology Integration Institute Summer Training*.

Queally, N. Introduction to airborne imaging spectroscopy: Topographic and BRDF corrections. (2023). *NSF Biology Integration Institute Summer Training*.

Mentoring

Hudson Billock	Undergraduate, UCLA	2025-2026
Nikita Burger	Undergraduate, UCLA	2025-2026
Bec Chenoweth	Undergraduate, UCLA	2025
Nate Trux	Undergraduate, UCLA	2024-2026
April Martinez	Undergraduate, UCLA	2024-2026
Cecilia Vanden Heuvel	Undergraduate, UW-Madison	2023-2024

Presentations and Posters (mentee, *invited)

Queally, N., Anderson, C.B., DeRanek, C., & Ordway, E. Drought, wildfire, and tree mortality: A decade of forest height trends in California (poster). (2026). *Ecological Society of America Annual*

Meeting.

Burger, N., **Queally, N.**, & Ordway, E. Variation of the foliar trait landscape across the chronogeographic continuum of the Central California coast (poster). (2026). *UCLA Ecology and Evolutionary Biology (EEB) Research Day*.

Martinez, A., **Queally, N.**, & Ordway, E. 2020 California wildfires: Comparing pre- and post-fire height and NDVI to track recovery (poster). (2026). *UCLA EEB Research Day*.

Billock, H., **Queally, N.**, & Ordway, E. Expanding hyperspectral CWC estimates over California using deep learning methods (poster). (2026). *UCLA EEB Research Day*.

Queally, N., Anderson, C.B., DeRanek, C., & Ordway, E. Drought, wildfire, and tree mortality: A decade of forest height trends in California (poster). (2025). *American Geophysical Union Fall Meeting*.

McMahon, C., **Queally, N.** SHIFTing from flora to fauna: Combining bioacoustics and imaging spectroscopy to track seasonal and spatial patterns of riparian animal vocal activity (poster). (2025). *American Geophysical Union Fall Meeting*.

***Queally, N.**, Johny, M., Zheng, T., Chadwick, K.D., Townsend, P.A. High frequency imaging spectroscopy data from SHIFT show phenological changes in trait expression in grazed landscapes. (2025). *Point Conception Institute & La Kretz Research Center Joint Science Symposium*.

Trux, N., **Queally, N.**, DeRanek, C., Zapata, F., Ordway, E. Evolution from the sky: Assessing the phylogenetic signal of Southern California plant communities from airborne VSWIR imagery (poster). (2025). *UCLA EEB Research Day*.

Vanden Heuvel, C., **Queally, N.**, Zheng, T., Berman, L., Townsend, P.A. Characterizing Forest and functional trait response to wildfire and climate extremes in the Sierra Nevada (poster). (2024). *Sierra Nevada Science Symposium*.

Queally, N., Johny, M., Brodrick, P.G., Townsend, P.A. High frequency imaging spectroscopy data from SHIFT show phenological changes in trait expression in grazed landscapes (poster). (2023). *American Geophysical Union Fall Meeting*.

Queally, N. Detecting disturbance legacy effects in functional trait phenology using imaging spectroscopy. (2023). *NASA Biodiversity and Ecological Conservation Team Meeting*.

Queally, N., Zheng, T., Ye, Z., Kovach, K., Schneider, F., Pavlick, R. P., & Townsend, P. A. Trade-offs in functional investments and tree dieback during the 2012 – 2016 California drought. (2023). *Ecological Society of America Annual Meeting*.

Queally, N., Zheng, T., Ye, Z., Kovach, K., Schneider, F., Pavlick, R. P., & Townsend, P. A. Trade-offs in functional investments influence tree mortality during 2012 – 2016 California drought (poster). (2022). *American Geophysical Union Fall Meeting*.

***Queally, N.**, Schimel, D., Turner, W., Reynolds, M., Davis, F., Miner, K., Chadwick, K. D., Gierach, M., Townsend, P. A., Pavlick, R. P., Brodrick, P. G. Imaging spectroscopy applications with SHIFT: SBG High Frequency Times Series. (2022). *Point Conception Institute Initiatives Workshop*.

Queally, N., Ye, Z., Kovach, K., Pavlick, R.P., Schneider, F., Townsend, P.A. BRDF correction of ABoVE AVIRIS-NG flightlines using FlexBRDF. (2021). *ABoVE Science Team Meeting* (online).

Queally, N., Ye, Z., Zheng, T., Chlus, A., Schneider, F., Pavlick, R.P., & Townsend, P.A. Automated BRDF Correction of Multiple Adjacent Flightlines in Complex Landscapes. (2020). *American Geophysical Union Fall Meeting* (online).

Queally, N., Foster, K., & Nickmeyer, A. California drought implications: Analyzing AVIRIS fraction of alive vegetation cover and climatic trends. (2017). *HyspIRI Mission Concept Science and Applications Workshop*.

Activities and Service

Service to the profession

Peer review: Remote Sensing of Environment, Applications in Plant Sciences, Bulletin of the American Meteorological Society (BAMS), IEEE Transactions on Image Processing (TIP), IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS), Nature Communications Earth & Environment, Earth's Future, New Phytologist

Invited review panelist: NASA 2023

Conference service:

Convener, *From Field to Space: Integrating Observations Across Scales for Terrestrial and Aquatic Ecosystem Monitoring*. American Geophysical Union Fall Meeting, San Francisco, CA, December 2026.

Service to the university

Forest ecologist faculty search committee member 2023

Graduate student representative 2019 – 2021

University of California Education Abroad Program scholarship reader 2018 – Present

Service to the community

Bear Divide Banding Station – Fecal sample technician 2026

LA Raptor Study – Nest monitor 2025

eBird – Data contributor 2019 – Present

iNaturalist – Data contributor 2019 – Present

Professional Engagement

Member, Ecological Society of America (ESA) 2023 – Present

Member, NSF Biology Integration Institute (BII): Advancing Spectral biology in Changing ENvironmnets to understand Diversity (ASCEND) 2021 – 2025

Member, American Geophysical Union (AGU) 2020 – Present